Optics

Optics is the branch of physics which involves the behavior and properties of light, including its interactions with matter and the construction of instruments that use or detect it. Optics usually describes the behavior of visible, ultraviolet, and infrared light.

<u>Newton's corpuscular theory of light</u>

Newton's corpuscular theory of light is based on the following points

1. Light consists of very tiny particles known as "corpuscular".

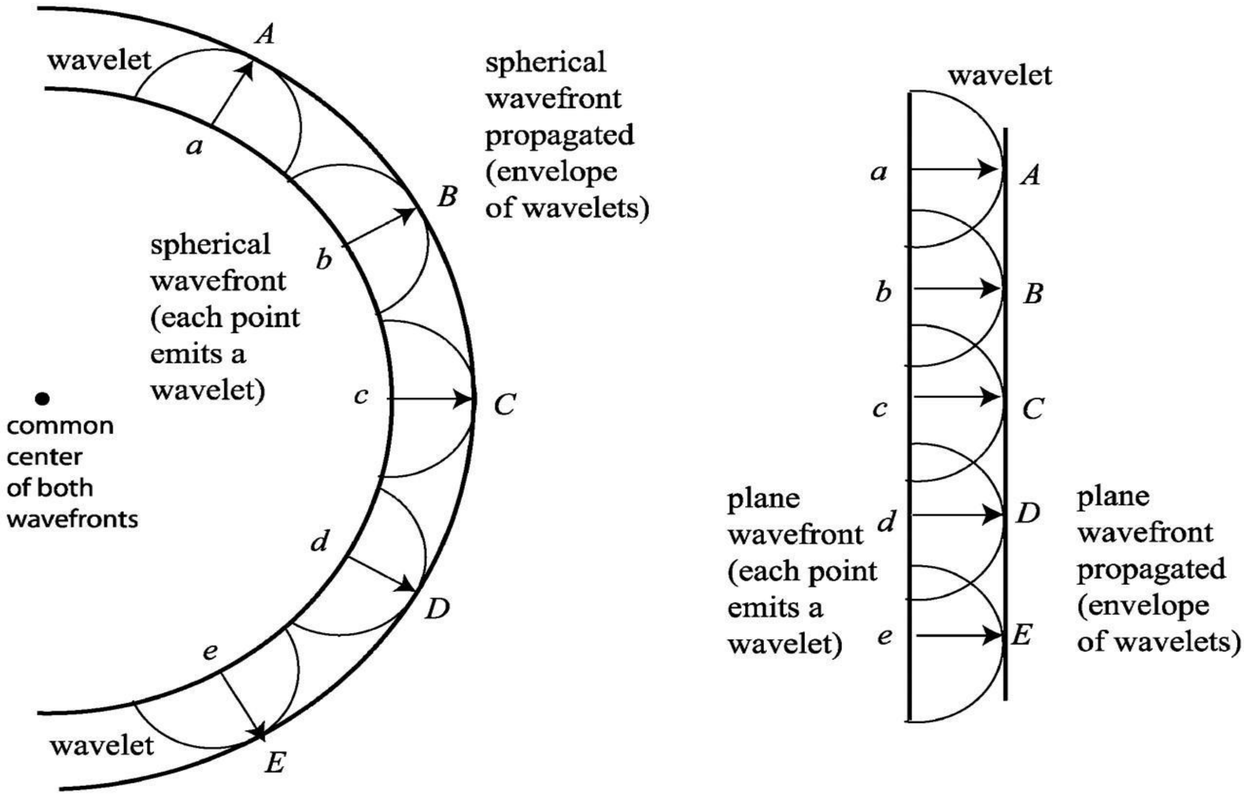
2. These corpuscles on emission from the source of light travel in straight line with high velocity.

3. When these particles enter the eyes, they produce image of the object or sensation of vision.

4. Corpuscles of different colors have different sizes.

<u>Huygen's Principle</u>

"Every point on a wavefront may be considered a source of secondary spherical wavelets which spread out in the forward direction at the speed of light. The new wavefront is the tangential surface to all of these secondary wavelets".



Superposition of waves

When a medium is disturbed simultaneously by any number of waves, the instantaneous resultant displacement of the media at every point at any instant is the algebraic sum of the displacements of the medium due to individual waves. In Young's own words, "When two undulations from different origin coincide either perfectly or very nearly, in direction, their joint effect is a combination of the motions belonging to each.'' Let us consider two waves crossed a particle in a medium and the different displacements of particle by the two waves are y_1 and y_2. If the waves are in same phase, then the resultant displacement of the particle is y = $y_1 + y_2$. And if they are in opposite phase, then the resultant displacement of the particle is y = $y_1 - y_2$.

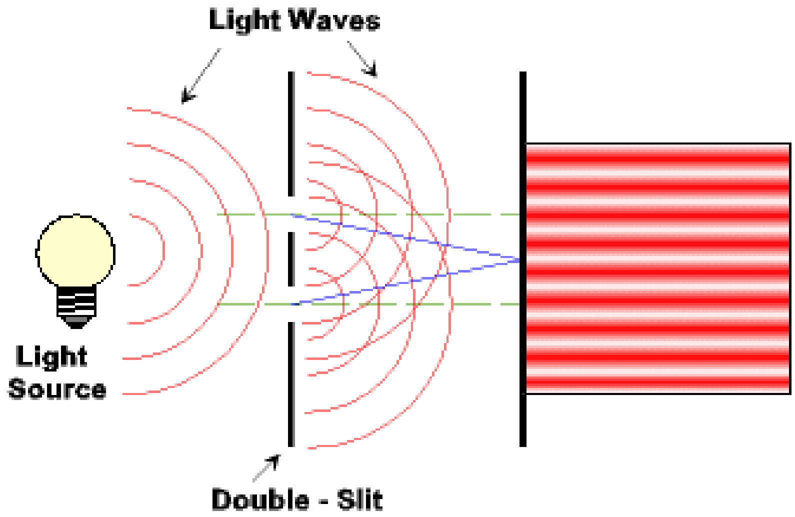
Coherent source

Two sources are said to be coherent if they emit light waves of exactly same frequency, and have zero or constant phase difference. Most of the light sources around us - lamp, sun, candle etc. are combination of multitude of incoherent sources of light. Laser is a coherent source i.e. constituent multiple sources inside the laser are phase-locked.

We need coherent sources of light in order to observe the effects of certain optical phenomena like Interference in lab. Two parallel slits lighted by a laser beam behind can be said to be two coherent point sources.

Interference of Light

In physics, **interference** is a phenomenon in which two waves superpose to form a resultant wave of greater or lower amplitude. Interference usually refers to the interaction of waves that are correlated or coherent with each other, either because they come from the same source or because they have the same or nearly the same frequency. Interference effects can be observed with all types of waves, for example, light, radio, acoustic, surface water waves or matter waves.



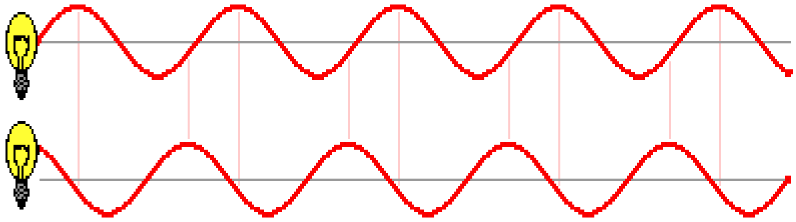
Conditions of Interference

When waves come together, they can interfere constructively or destructively. To set up a stable and clear interference pattern, two conditions must be met:

1. The sources of the waves must be coherent, which means they emit identical waves with a constant phase difference.
2. The waves should be monochromatic - they should be of a single wavelength.

Coherent Sources

Those sources of light which emit light waves continuously of same wavelength, time period, frequency, and amplitude and have zero phase difference or constant phase difference are called coherent sources.



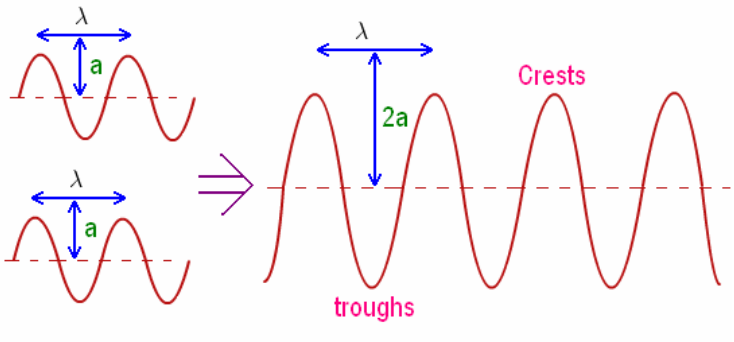
These two waves are coherent - they have a phase difference which is constant over time.

There are two types of interference.

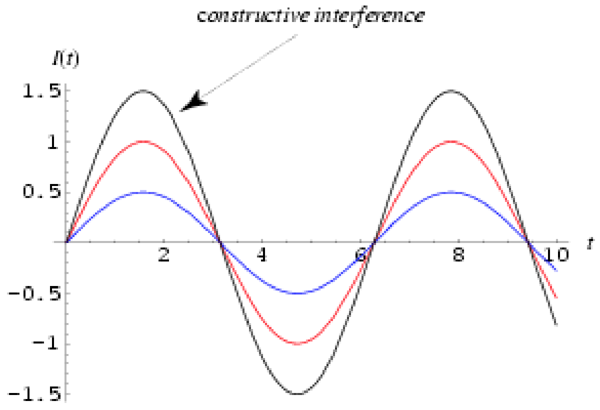
1. Constructive Interference.
2. Destructive Interference.

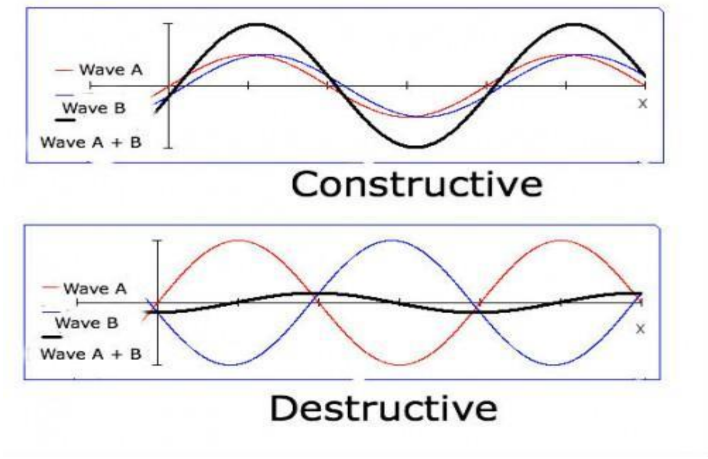
Constructive Interference

When two light waves superpose with each other in such a way that the crest of one wave falls on the crest of the second wave, and trough of one wave falls on the trough of the second wave, then the resultant wave has larger amplitude and it is called constructive interference.



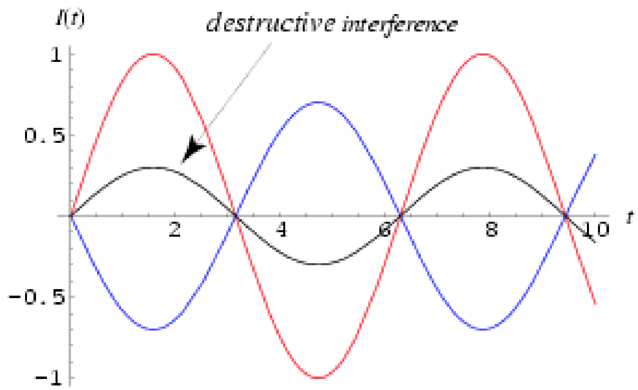
Crests and trough reinforce

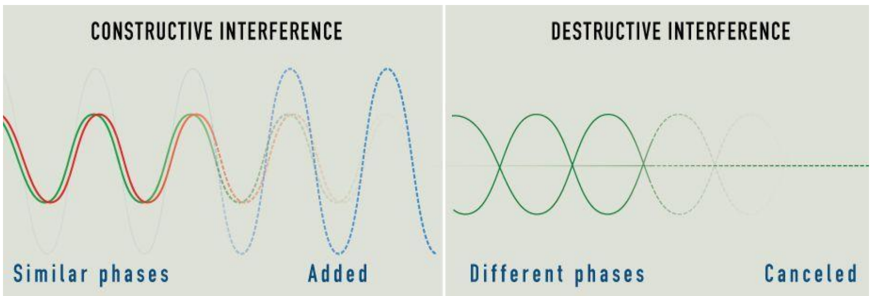




Destructive Interference

When two light waves superpose with each other in such a way that the crest of one wave coincides the trough of the second wave, then the amplitude of resultant wave becomes zero and it is called destructive interference.





Young's Double Slit Experiment

In 1801 an English physicist named Thomas Young performed an experiment that strongly inferred the wave-like nature of light.

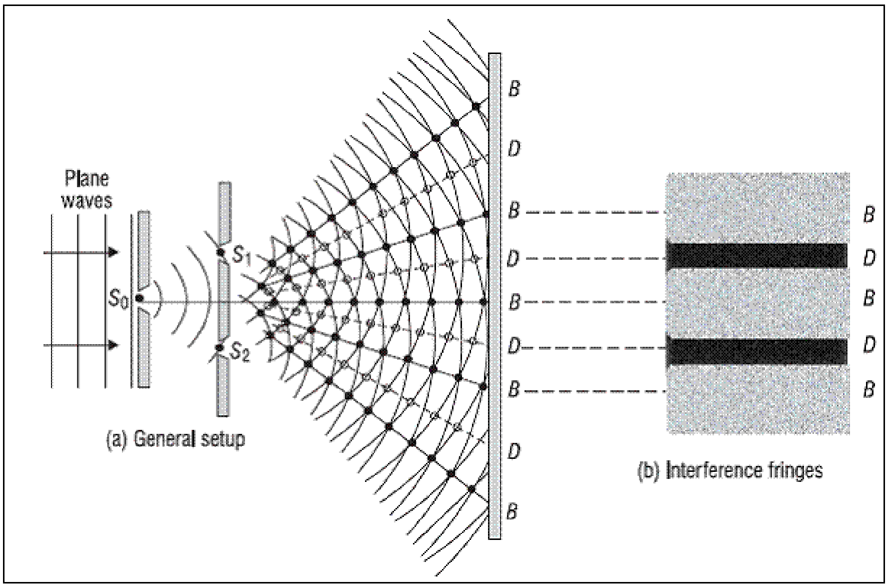


Fig: Young's double slit experiment

Thomas Young demonstrated the experiment on the interference of light. He allowed light to fall on a pinhole S_0 and then at some distance away on two pinholes S_1 and S_2. S_1 and S_2 are equidistant from S_0 and are close to each other. Spherical waves spread out from S_0. Spherical waves also spread out from S_1 and S_2. These waves are of the same amplitude and wavelength. On the screen, interference bands are produced which are alternatively dark and bright. The points such as B are bright because the crest due to the one wave coincides with the rest due to the other and therefore, they reinforce with each other. The points such as D are dark because of the crest of one fall to on the trough of the other and they neutralize the effect of each other. Points, similar to B, where the trough of one fall on the trough of the other, are also bright because the two waves reinforce.

It is not possible to show interference due to two independent sources of light, because a large number of difficulties are evolved. The two sources may emit light waves of largely different amplitude and wavelength and the phase difference between the two may change with time.

Theory of Interference Fringes

Let us consider a narrow monochromatic source S and two pinholes S_1 and S_2 equidistant from S. S_1 and S_2 acts as two coherent sources separated by a distance d.

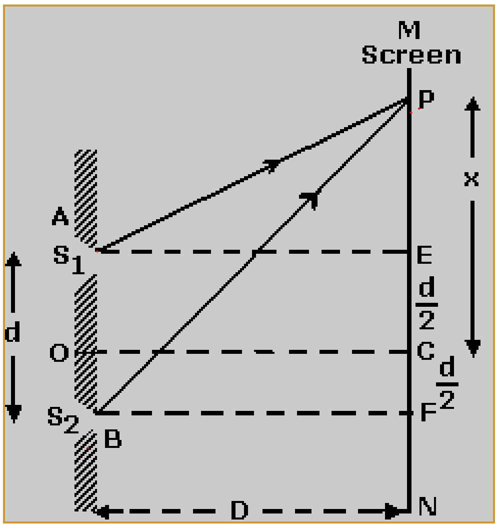


Fig: Young's double slit experiment

Let a screen be placed at a distance D from the coherent source. The point C on the screen is equidistant from S_1 and S_2. Therefore, the path difference between the two waves is zero. Thus, the point C has maximum intensity.

Let us consider a point P at a distance x from C. The waves reach at the point P from A and B.

Here,

$$PE = x - \frac{d}{2}$$

$$PF = x + \frac{d}{2}$$

Now from the triangle PAE we get,

$$(AP)^2 = D^2 + \left(x - \frac{d}{2}\right)^2 \text{ ------------------------(1)}$$

Similarly, from the triangle PBF we get

$$(BP)^2 = D^2 + \left(x + \frac{d}{2}\right)^2 \text{ ------------------------(2)}$$

Subtracting equation (1) from equation (2)

$$(BP)^2 - (AP)^2 = \left[D^2 + \left(x + \frac{d}{2}\right)^2\right] - \left[D^2 + \left(x - \frac{d}{2}\right)^2\right]$$

$$or, (BP)^2 - (AP)^2 = D^2 + x^2 + xd + \frac{d^2}{4} - D^2 - x^2 + xd - \frac{d^2}{4}$$

$$or,\ (BP)^2 - (AP)^2 = 2xd$$

$$or, (BP - AP)(BP + AP) = 2xd$$

$$or, (BP - AP) = 2xd/(BP + AP) \text{------------------------(3)}$$

As, $d \ll D$. So, we can assume that $BP \approx AP \approx D$. So, from equation (3) we can write-

$$(BP - AP) = 2xd/(D + D)$$

$$or, (BP - AP) = 2xd/(D + D)$$

$$\therefore (BP - AP) = xd/D$$

But $(BP - AP)$ is the path difference of the light at point P.

$$\text{The path difference} = xd/D \text{ ----------------------------(4)}$$

Again, we know, the phase difference $= \frac{2\pi}{\lambda} \times$ the path difference.

$$\text{So, the phase difference of the light at point P} = \frac{2\pi}{\lambda} \times \frac{xd}{D} \text{ -----------------(5)}$$

Condition for bright fringes:

If the path difference is a whole number multiple of wavelength λ, then the points will be bright.

i.e. $\frac{x\,d}{D} = n\,\lambda$ Where n = 0, 1, 2, 3……………..

$$\text{or, } x = \frac{n\,\lambda\,D}{d} \text{-------------------(6)}$$

This equation gives the distances of the bright fringes from the point C. At C, the path difference is zero and a bright fringe is formed.

$$\text{When } n = 0,\ \ x_0 = 0$$

$$n = 1,\ \ x_1 = \frac{\lambda\,D}{d}$$

$$n = 2,\ \ x_2 = \frac{2\,\lambda\,D}{d}$$

$$n = 3,\ \ x_3 = \frac{3\,\lambda\,D}{d}$$

………………………………

$$n = n,\ \ x_n = \frac{n\,\lambda\,D}{d}$$

Therefore, the distance between any two consecutive bright fringes

$$x_2 - x_1 = \frac{2\,\lambda\,D}{d} - \frac{\lambda\,D}{d} = \frac{\lambda\,D}{d} \text{ -------------------(7)}$$

Condition for dark fringes:

If the path difference is an odd number multiple of half wavelength, then the point will be dark.

$$\text{i.e., } \frac{x\,d}{D} = \frac{2n+1}{2}\lambda$$

$$4\ x = \frac{(2n+1)\,\lambda\,D}{2d}$$

This equation gives the distances of the dark fringes from the point C.

$$\text{When, } n = 0,\ \ x_0 = \frac{\lambda\,D}{2\,d}$$

$$n = 1,\ \ x_1 = \frac{3\,\lambda\,D}{2\,d}$$

$$n = 2,\ \ x_2 = \frac{5\lambda\,D}{2d}$$

$$n = 3,\ \ x_3 = \frac{7\,\lambda\,D}{2\,d}$$

………………………………

$$n = n,\ \ x_n = \frac{(2n+1)\,\lambda\,D}{2d}$$

The distance between any two consecutive dark fringes,

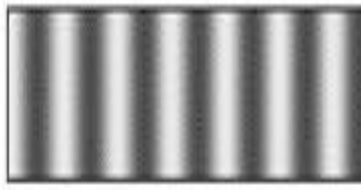
$$x_2 - x_1 = \frac{5\,\lambda\,D}{2d} - \frac{3\lambda\,D}{2d} = \frac{\lambda\,D}{d} \quad \text{---------------------------(8)}$$

The distance between any two-consecutive bright or dark fringes is known as fringe width.

Therefore, alternately bright and dark parallel fringes are formed. The fringes are formed on both sides of C. Moreover, from equations (7) and (8), it is clear that the width of the bright fringe is equal to the width of the dark fringe. All the fringes are equal in width and are independent of the order of the fringe. The breadth of a bright or a dark fringe is, however, equal to half the fringe width and is equal to $\frac{\lambda D}{2d}$.

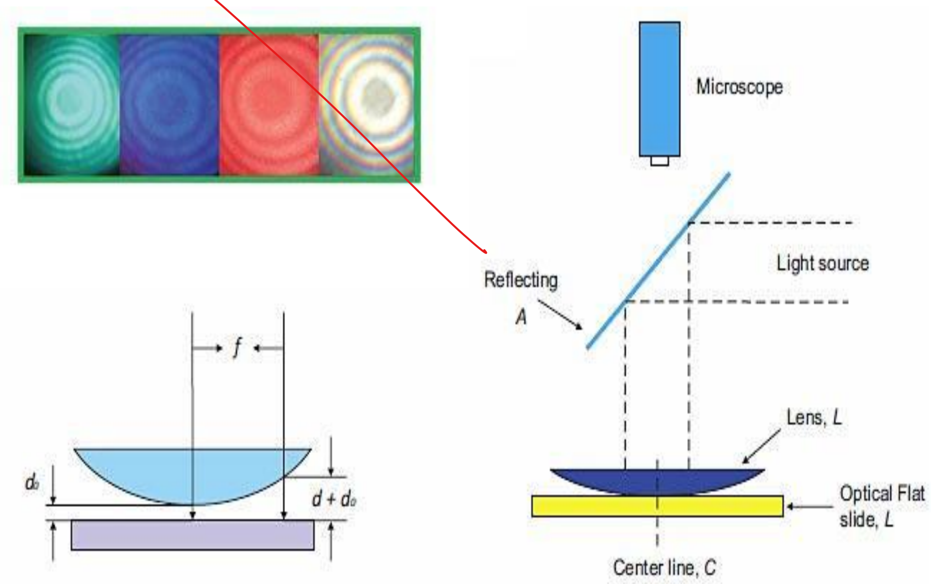
$$\text{The fringe width} = \frac{\lambda D}{d}$$

Therefore, (i) the width of the fringe is directly proportional to the wavelength of light, $\propto \lambda$. (ii) The width of the fringe is directly proportional to the distance between the two sources, $\propto \frac{1}{d}$. Thus, the width of the fringe increases (a) with increase in wavelength (b) with increase in the distance D and (c) by bringing the two sources A and B close to each other.



Newton’s Rings

Newton's ring is a phenomenon in which an interference pattern is created by the reflection of light between two surfaces—a spherical surface and an adjacent flat surface. It is named after Isaac Newton, who first studied them in 1717.



Write down a short note on Newton’s Rings Experiment

When a Plano-convex lens of long focal length is placed on a plane glass plate, a thin film of air is enclosed between the lower surface of the lens and the upper surface of the plate. The thickness of the air film is very small at the point of contact and gradually increases from the center outwards. The fringes produced with monochromatic light are circular. The fringes are concentric circles, uniform in thickness and with the point of contact as the center. When viewed with white light, the fringes are colored. With monochromatic light, bright and dark circular fringes are produced in the air film.

The arrangement used is shown in the figure below. S is a source of sodium light. A parallel beam of light from lens L_1 is reflected by the glass plate G inclined at an angle of 45^0 to the horizontal. L is a plano-convex lens of large focal length. Newton's rings are viewed through G by the travelling microscope M focused on the air film. Circular bright and dark rings are seen with the center dark. With the help of travelling microscope, the diameter of the n^{th} ring can be measured.

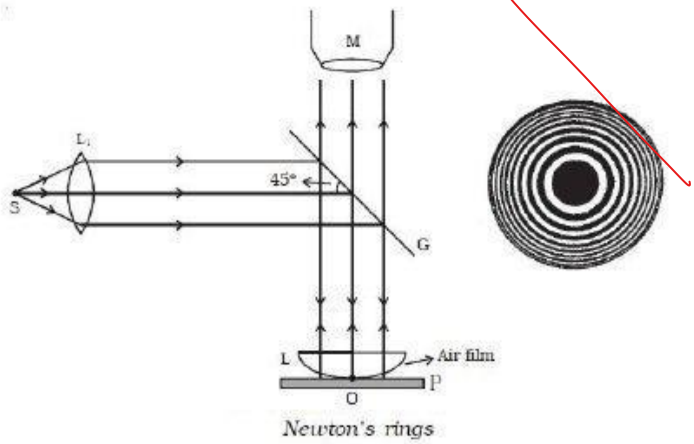
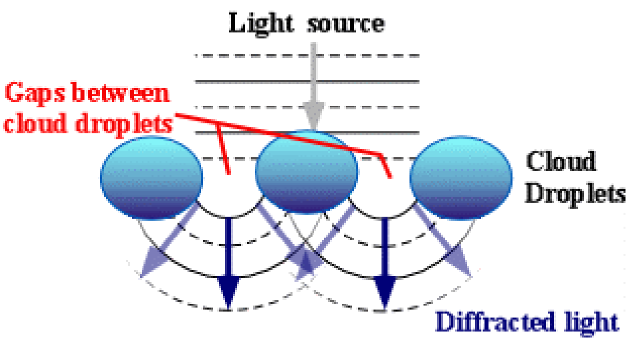


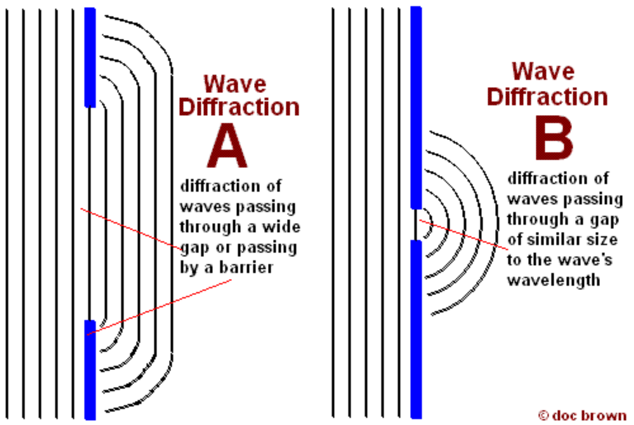
Fig.: Newton's ring experiment

Diffraction

What do you mean by diffraction of light?

Diffraction is the slight bending of light as it passes around the edge of an object. The amount of bending depends on the relative size of the wavelength of light to the size of the opening. If the opening is much larger than the light's wavelength, the bending will be almost unnoticeable. However, if the two are closer in size or equal, the amount of bending is considerable, and easily seen with the open eye.





What are the conditions of diffraction of light?

There are two conditions for the production of diffraction

(1) In case of straight edge: The edge should be very sharp and its width is to be equal to or is of the order of the wavelength, λ of light.

(2) In case of thin hole: the diameter of the hole should be extremely very small such that it is equal to or is of the order of the wavelength λ of light.

What are types of diffraction? Define them.

Diffraction is of two types

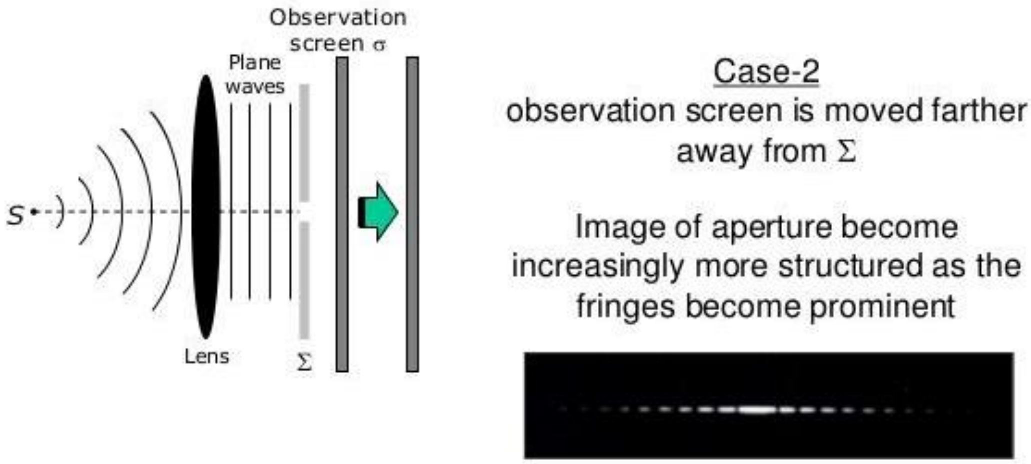
(1) Fresnel's class of diffraction

(2) Fraunhofer's class of diffraction

Fresnel's class of diffraction

When the source of light and the screen are at finite distance from the inside obstacle, then the diffraction observed due to the obstacle is called the Fresnel class of diffraction.

In this type of diffraction wave fronts are generally spherical or cylindrical. This type of diffraction occurs in a straight edge, fine wire and narrow slit.

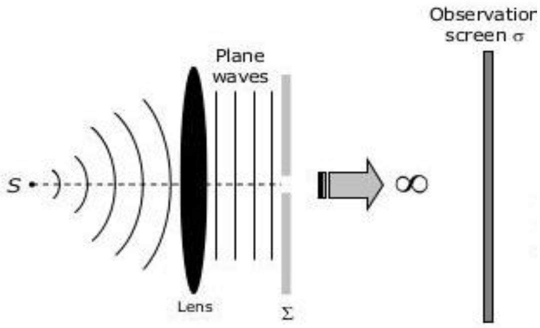


Fresnel or Near-Field Diffraction

Fraunhofer's class of diffraction

When the source of light and the screen are effectively at infinite distance from the obstacle or aperture causing diffraction, then that type of diffraction is called Fraunhofer's class of diffraction.

In this type of diffraction wave fronts incident on the obstacle or aperture is plane.

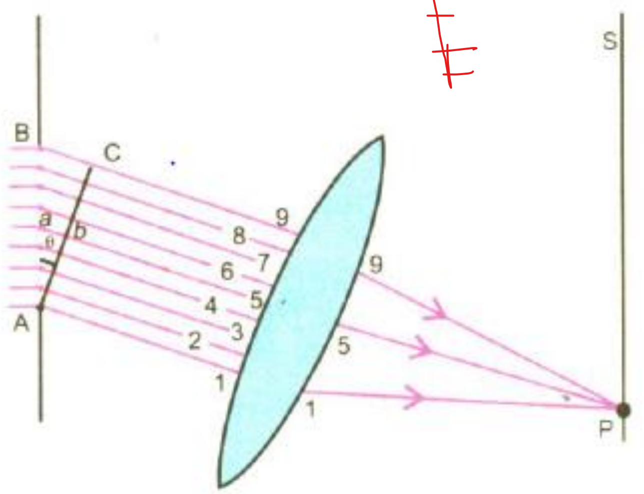


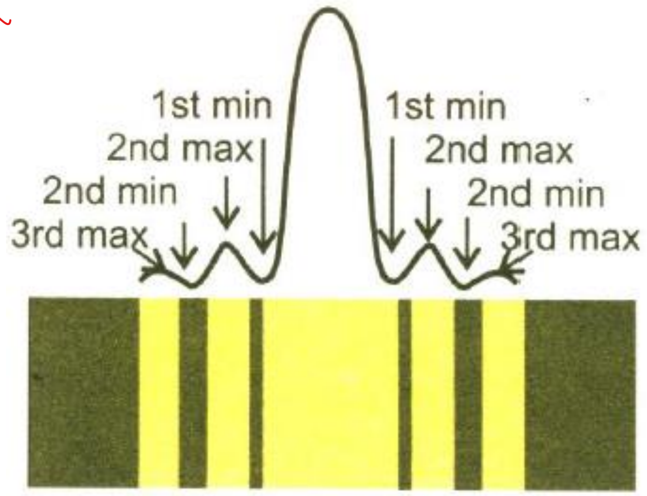
Fraunhofer or Far-Field Diffraction

Explain diffraction of monochromatic light due to a narrow slit

The experimental arrangement for studying diffraction of light due to a narrow slit is shown in the figure.

The slit AB of width d is illuminated by a parallel beam of monochromatic light of wavelength λ. The screen S is placed parallel to the slit AB. Rays of light are brought to focus on the screen. A small portion of the incident wavefront passes through the narrow slit. Each point of wavefront sends out secondary wavelets to the screen. These wavelets then interfere to produce diffraction. We take rays instead of wavefronts to show diffraction on the screen. In figure, only 9 rays have been shown. Actually, there are large numbers of rays.





Let us consider two rays 1 and 5 which are in phase when they are at wavefront AB. When these reach the wavefront AC, ray 5 should have a path difference ab.

In triangle ΔaAb

$$\sin\theta = \frac{ab}{Aa} = \frac{ab}{\frac{d}{2}}$$

$$» \ \frac{d}{2}\sin\theta = ab$$

So the path difference $ab = \frac{d}{2}\sin\theta$

For first minima (destructive interference),

Path difference $= \frac{\lambda}{2}$

$$\text{Therefore, } \frac{\lambda}{2} = \frac{d}{2}\sin\theta$$

$$d \sin\theta = \lambda$$

In general, the condition for different orders of minima on either side of center is given by:

$$d \sin\theta = n\lambda$$

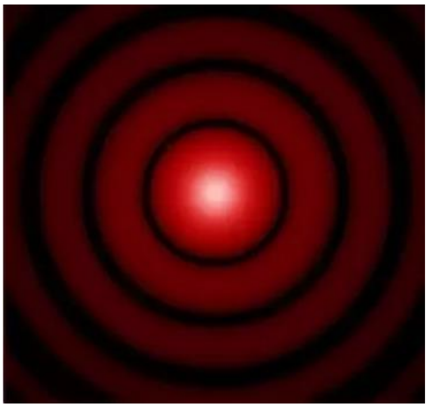
where n= ±(1,2,3,……)

Define Diffraction grating?

A diffraction grating is glass plate having a large number of close parallel equidistant slits mechanically ruled on it. The transparent spacing between the scratches on the glass plate act as slits.

Distinguish between interference and diffraction of light

1. Two separate wave fronts originating from two coherent sources produce interference. Secondary wavelets originating from different parts of the same wave front constitute diffraction.
2. The region of minimum intensity is perfectly dark in interference. In diffraction they are not perfectly dark.
3. Width of the fringes is equal in interference. In diffraction they are never equal.
4. The intensity of all positions of maxima are of the same intensity in interference. In diffraction they do vary.
5. When we have two infinitely narrow slits separated by a distance apart near the source, we get interference. But when we have a single slit of finite width or rather an aperture near the source, we get diffraction.



Polarization

What do you mean by Polarization?

In an electromagnetic wave, both the electric field and magnetic field are oscillating but in different directions. The process by which light waves vibrating in different planes can be made to vibrate in a particular plane is called polarization of light.

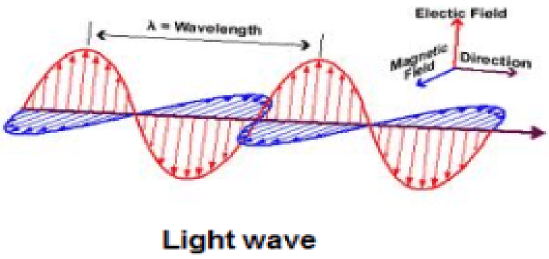
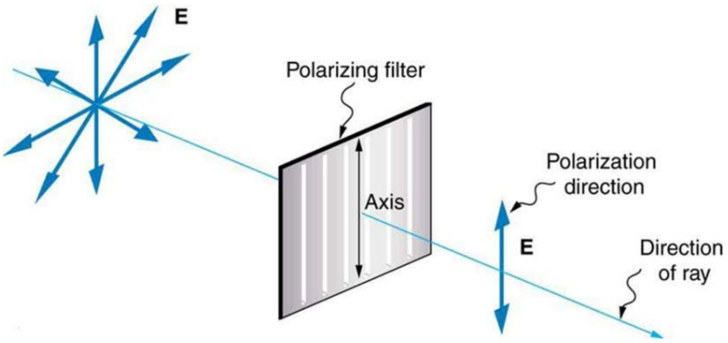
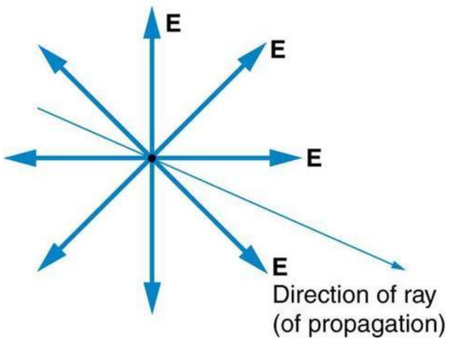


Fig. 1

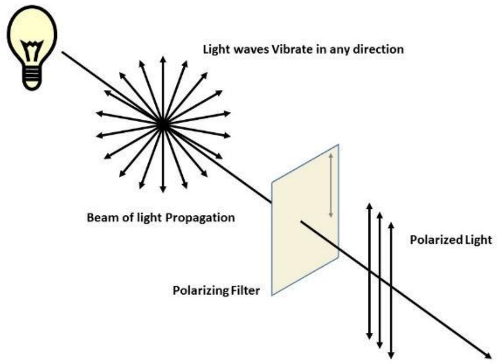


Define Unpolarized light, Polarized light, Plane of vibration, Plane of polarization

(1) **Unpolarized light:** Ordinary light waves that vibrate normal to the direction of propagation, spread all around the source with equal amplitude is called unpolarized light. The figure given below shows the unpolarized light.

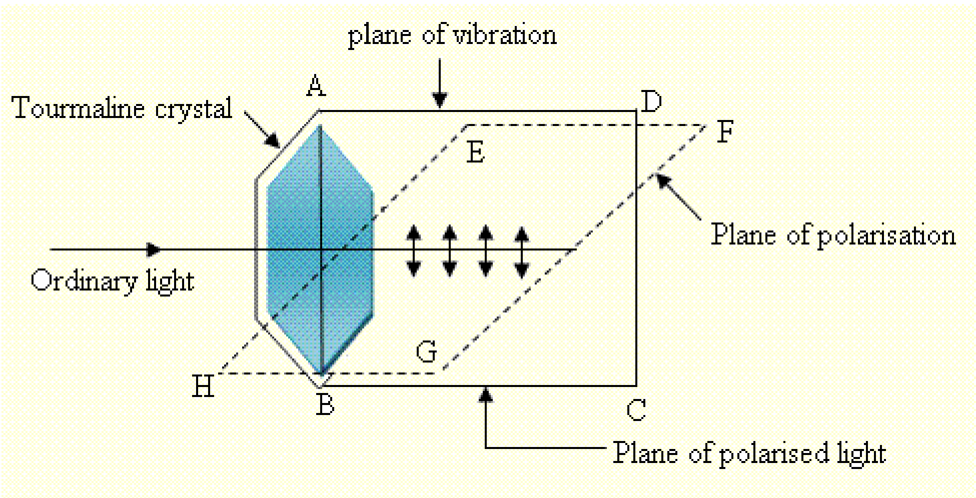


(2) Polarized light: The transverse light waves vibrating on a particular plane or parallel to it is called polarized light.



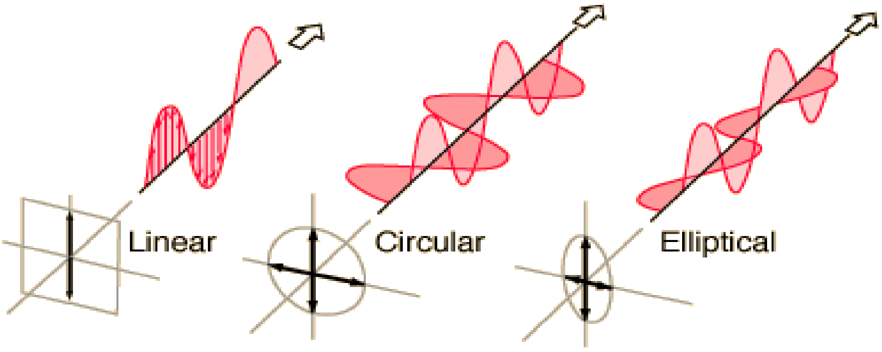
(3) Plane of vibration: The plane in which the particles of the light waves vibrate is called the plane of vibration. The below figure shows the plane of vibration. ABCD is the plane of vibration.

(4) Plane of polarization: The plane which exists normal to the plane of vibration is called the plane of polarization. In the figure EFGH is the plane of polarization.



<u>What are the Classifications of Polarization?</u>

1. Linear Polarization
2. Circular Polarization
3. Elliptical Polarization



Write down the optical processes by which linearly polarized light can be produced

Linearly polarized light may be produced from unpolarized light using of the following five optical phenomena:

1. Reflection
2. Refraction
3. Scattering
4. Selective absorption (dichroism) and
5. Double refraction

State Brewster's Law

In 1811 Brewster's proposed it. The law states that the tangent of the angle at which polarization is obtained by reflection is numerically equal to the refractive index of the medium.

If θ_p is the angle and μ is the refractive index of the medium, then

$$\mu = tan\ tan\ \theta_p$$

This is known as Brewster's law.

Prove Brewster's law Mathematically

If natural light is incident on a smooth surface at the polarizing angle, it is reflected along PC and refracted along PB, as shown in figure. Brewster found that the maximum polarization of reflected ray occurs when it is at right angle to the refracted ray.

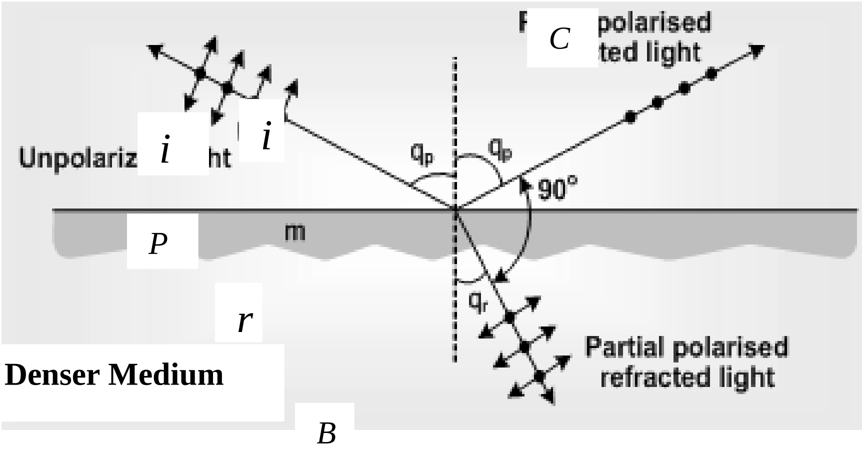


Fig.: Production of plane polarized light by the method of reflection.

It means that

$$i + r = 90°$$

$$\therefore r = 90° - i \text{ ---------------------------------(1)}$$

Where, $i \text{ and } r$ represents the incident angle and the refractive angle respectively.

According the Snell's law,

$$\frac{\sin\sin i}{\sin\sin r} = \frac{\mu_2}{\mu_1} \qquad \text{-----------------------------------(2)}$$

Where μ_2 and μ_1 are the absolute refractive index of the reflecting surface and the surrounding medium.

It follows from equation (1) and (2) that

$$\frac{\sin\sin i}{\sin\sin(90° - i)} = \frac{\mu_2}{\mu_1}$$

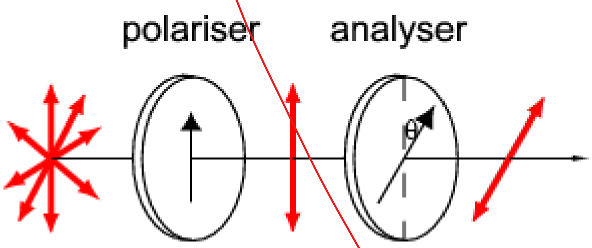
$$or, \frac{\sin\sin i}{\cos\cos i} = \frac{\mu_2}{\mu_1}$$

$$\therefore \tan\tan i = \frac{\mu_2}{\mu_1} \qquad \text{---------------------------------(3)}$$

Equation (3) shows that the polarizing angle depends on the refractive index of the reflecting surface. The polarizing angle i is also known as Brewster angle and denote as θ_B. Light reflected from any angle other than Brewster angle is partially polarized.

<u># State Malus Law and explain it mathematically.</u>

According to Malus, when completely plane polarized light is incident on the analyzer, the intensity E_1 of the light transmitted by the analyzer is directly proportional to the square of the cosine of angle between the transmission axes of the analyzer and the polarizer.



If E_1 is the intensity of the transmitted wave and θ is the angle between the planes of polarizer and the analyser. Then,

$$E_1 \propto \theta$$

$$E_1 = E\theta \quad ;$$

Where E is the intensity of the incident polarized light.

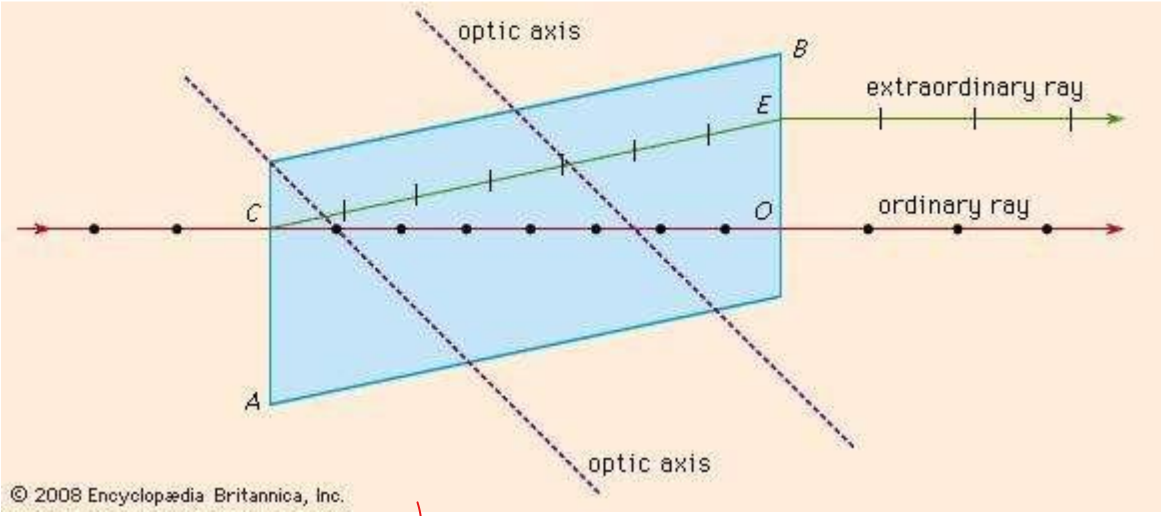
This is known as Malus law of polarization.

What do you mean by Double Refraction?

Double refraction, also called **birefringence**, an optical property in which a single ray of unpolarized light entering an anisotropic medium is split into two rays, each traveling in a different direction. One ray (called the extraordinary ray) is bent, or refracted, at an angle as it travels through the medium; the other ray (called the ordinary ray) passes through the medium unchanged.

Double refraction can be observed by comparing two materials, glass and calcite ($CaC0_3$). If a pencil mark is drawn upon a sheet of paper and then covered with a piece of glass, only one image will be seen; but if the same paper is covered with a piece of calcite, and the crystal is oriented in a specific direction, then two marks will become visible.

The Figure shows the phenomenon of double refraction through a calcite crystal. An incident ray is seen to split into the ordinary ray *CO* and the extraordinary ray *CE* upon entering the crystal face at *C*. If the incident ray enters the crystal along the direction of its optic axis, however, the light ray will not become divided.



What do you mean by Optical rotation?/ Write a short note on Optical rotation.

When plane polarized light is passed through certain crystals, they rotate the plane of polarization. This phenomenon is called optical rotation. The crystals which show this phenomenon are called optically active crystals. Quartz and sodium chloride are the example of optically active crystals.

Certain solutions of organic substance, such as sugar and tartaric acid, show optical rotation. This property of optically active substances can be used to determine their concentration in the solutions.

What do you mean by Polarimeters?

A **polarimeter** is a scientific instrument used to measure the angle of rotation caused by passing polarized light through an optically active substance.

Some chemical substances are optically active, and polarized (uni-directional) light will rotate either to the left (counter-clockwise) or right (clockwise) when passed through these substances. The amount, by which the light is rotated, is known as the angle of rotation. The angle of rotation is basically known as observed angle.

What do you mean by resolving power of an optical instrument?

The ability of an optical instrument, expressed in numerical measure, to distinguish between the images of two nearby objects is termed as its resolving power. In case of a diffraction grating, resolving power is referred to as its ability to resolve two spectral lines whose wavelengths are very close to each other so that these two lines can be viewed as separate lines.